



Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

- VERIFIED 0.80 - 1.00** 3+ independent sources including media
- CONFIRMED 0.50 - 0.79** 2 sources or 1 high-quality media
- EMERGING 0.25 - 0.49** Single-source; passes AI keyword filter

## Source Breakdown

| Source          | Articles | Category  |
|-----------------|----------|-----------|
| ArXiv CS.AI     | 134      | Media     |
| TechCrunch AI   | 10       | Media     |
| HackerNews      | 10       | Community |
| The Verge AI    | 5        | Media     |
| MIT Tech Review | 2        | Media     |
| VentureBeat AI  | 1        | Media     |

## VERIFIED INTELLIGENCE — 4 stories

VERI

**Claude's voice mode is now available for Opus and Sonnet**

The Verge AI

+2 sources

100%

Until now, voice mode has only been available on Claude Haiku, Anthropic's faster but less powerful model. Now the company is making its Opus and Sonnet models available in voice mode, and extending its reach into apps like Gmail, Slack, and Canva. When Anthropic launched voice...

<https://www.theverge.com/ai-artificial-intelligence/970065/anthropic-voice-mode-claude-...>

VERI

**Show HN: Verify what an AI agent did, then tamper with the record (no signup)**

HackerNews

+1 source

90%

HN 2

This is today's edition of The Download, our weekday newsletter that provides a daily dose of what's going on in the world of technology. China's AI models have Trump's AI world at war with itself Last weekend, several current and former advisers to President Donald Trump on AI...

<https://api.cinchor.com/proof/refund>

VERI

**OpenAI is making big claims as it rolls out ChatGPT Health to everyone**

The Verge AI

+1 source

80%

OpenAI is rolling out ChatGPT Health to everyone in the US on Thursday, allowing more people to connect their medical records and health-tracking information to the chatbot. During a briefing, Ashley Alexander, OpenAI's vice president of health product, says the company's models...

<https://www.theverge.com/ai-artificial-intelligence/970115/openai-chatgpt-health-launch...>

VERI

**Alexa Plus is getting an AI update to handle more complicated instructions**

The Verge AI

+1 source

80%

Amazon is launching an update to its Alexa Plus assistant that will allow it to connect to smart home devices in new ways. With the update, which is currently in preview, Alexa Plus can link up with tech from Bosch, Delta, Ecovacs, iRobot, Yale Home, Whirlpool, Tapo, Eufy, and...

<https://www.theverge.com/tech/970399/amazon-alexa-plus-ai-update-smart-home-devices>

## CONFIRMED SIGNALS — 153 stories

CONF

**Show HN: Palmier Pro - Open-source macOS video editor built for AI**

HackerNews

60%

HN 138

Hi HN, we are Marcos and Harrison, cofounders of Palmier ( <https://palmier.io> ). We are building Palmier Pro, an open source macOS video editor, with built-in AI generation and a local MCP server that connects to your agent. Here are a few demos: - Making some AI transitions:...

<https://github.com/palmier-io/palmier-pro>

CONF

**Show HN: Claude-thermos keeps your Claude session warm for you**

HackerNews

50%

HN 94

No summary — see link for full article.

<https://github.com/izeigerman/claude-thermos>

## CONF

**Show HN: OneCLI - OSS credential gateway that keeps secrets out of AI agents**

HackerNews 50% HN 88

hey HN, Jonathan and Guy here, creators of OneCLI ( <https://onecli.sh/> ). OneCLI is an open source vault for AI Agents. Traditional vaults are used to store your secrets and, on demand, provide them to you all in a secure way, trusting the person to keep them safe. We figured...

<https://github.com/onecli/onecli>

## CONF

**Show HN: Connecting my dumb garage and car's homelink buttons to Home Assistant**

HackerNews 50% HN 9

Hi all, I've been on parental leave and while working on my garage door, I thought to myself... it would be great to open up this garage door from my phone without having to buy a bunch of expensive stuff. That snowballed into, I want the garage door buttons on my car's rear...

<https://dsauerbrun.com/writing/dumb-garage>

## CONF

**Show HN: I simulated closing the Strait of Hormuz on real oil trade data**

HackerNews 50% HN 8

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<https://globaloilnetwork.staffinganalytics.io/>

## 0 CONF

**Show HN: What's AI's go-to, public or private healthcare?**

HackerNews 50% HN 8

No summary — see link for full article.

<https://www.modelbias.ai/prompt/public-or-private-healthcare>

## 1 CONF

**Show HN: A primer for non-coders to design software via structured AI dialogue**

HackerNews 50% HN 6

No summary — see link for full article.

<https://github.com/CoreGrowthLabs/CoreGrowthPrompting/blob/main/README.en.md>

## 2 CONF

**Show HN: Notebooker.ai - NotebookLM alternative, your own models, keys, storage**

HackerNews 50% HN 3

Hi HN. This is a personal side project I've been building for about six months on top of the open-source Open Notebook platform ( <https://github.com/Ifnovo/open-notebook> ). Notebooker saves the stuff you'd otherwise lose in tabs and bookmarks - links, PDFs, audio, video - and...

<https://notebooker.ai/>

## 3 CONF

**Show HN: 5dive - Run a Company of Claude Code/Codex Agents (Written in Bash)**

HackerNews 50% HN 3

No summary — see link for full article.

<https://github.com/5dive-ai/5dive>

## 4 CONF

**AMD takes on Nvidia with its Helios AI rack-scale system**

TechCrunch AI 50%

AMD is challenging its chipmaker rival with a new rack-scale system that will start shipping to customers later this year.

<https://techcrunch.com/2026/07/23/amd-takes-on-nvidia-with-its-helios-ai-rack-scale-sys...>

5 **CONF****AegisAI, founded by former Google security execs, lands \$36M to stop AI-driven spear phishing****TechCrunch AI** 50%

AegisAI co-founders developed AI agents that quickly analyze each message as a human would, paying attention to small anomalies that even the most elaborate checklist wouldn't catch.

<https://techcrunch.com/2026/07/23/aegisai-founded-by-former-google-security-execs-lands...>

6 **CONF****Runway launches AI model router as generative media gets crowded****TechCrunch AI** 50%

The Media Router is a tool that automatically selects the best image, video, or audio generation model for a request based on whether a developer prioritizes quality, speed or cost.

<https://techcrunch.com/2026/07/23/runway-bets-on-ai-model-routing-as-generative-media-g...>

7 **CONF****Meta launched a new AI optimism ad set to a song about human extinction****TechCrunch AI** 50%

David Bowie's song "Five Years," which Meta used in a supposedly inspiring advertisement, is about humans learning that they have five years left to live before the apocalypse.

<https://techcrunch.com/2026/07/23/meta-launched-a-new-ai-optimism-ad-set-to-a-song-abou...>

8 **CONF****AI chip startup Etched defies skeptics, hits \$10.3B valuation from big-name investors****TechCrunch AI** 50%

Etched, founded by three Harvard dropouts, has created new chips and memory components that speed up inference on any AI model -- no GPUs required, it says.

<https://techcrunch.com/2026/07/23/ai-chip-startup-etched-defies-skeptics-hits-10-3b-val...>

9 **CONF****Google's Gemini nears billion-user milestone****TechCrunch AI** 50%

Gemini had over 750 million monthly users in February.

<https://techcrunch.com/2026/07/23/google-closes-in-on-another-billion-user-product-with...>

0 **CONF****Experts say exploiting Anthropic's Fable isn't how Kimi K3 got so good****TechCrunch AI** 50%

"I don't think you get a model this strong and this quickly on the heels of Fable doing strictly distillation," one expert told TechCrunch.

<https://techcrunch.com/2026/07/23/experts-say-exploiting-anthropics-fable-isnt-how-kimi...>

1 **CONF****How OpenAI's human mistake led to the AI-powered hack on Hugging Face****TechCrunch AI** 50%

OpenAI made a mistake setting up what it called a "highly isolated" testing environment and sandbox. According to cybersecurity experts, that human mistake is what made the AI-powered attack on Hugging Face possible.

<https://techcrunch.com/2026/07/22/how-an-openais-human-mistake-led-to-the-ai-powered-ha...>

2 **CONF****How AI helps scientists design the next generation of medicines****MIT Tech Review** 50%

Designing and developing a new medicine is an expensive, failure-prone scientific challenge. A new drug can take many years to develop, at the cost of a significant investment. And even then, most possible candidates never reach the patient. For biologic medicines, therapies...

<https://www.technologyreview.com/2026/07/23/1140346/how-ai-helps-scientists-design-the-...>

3 CONF

**Lawmakers prepare bill requiring AI 'kill switch'**

The Verge AI 50%

Lawmakers are preparing to introduce an "AI Kill Switch Act" that would require AI companies to shut down or throttle their systems on orders from the Department of Homeland Security, according to a report from Politico. Reps. Ted Lieu (D-CA) and Nathaniel Moran (R-TX) are...

<https://www.theverge.com/ai-artificial-intelligence/969939/lawmakers-ai-kill-switch-pro...>

4 CONF

**Apple's OpenAI lawsuit is about who gets to define the post-smartphone era**

The Verge AI 50%

Today on Decoder, I'm talking with Hayden Field, The Verge's senior AI reporter, about the major trade secrets lawsuit between Apple and OpenAI and what this tells us about OpenAI's future. By now I'm sure most Decoder listeners are familiar with Apple's allegations in this...

<https://www.theverge.com/podcast/968787/apple-openai-trade-secrets-lawsuit-ai-hardware-...>

5 CONF

**AINTEMA: Agentic AI Architecture for Autonomous Test Management with Generative Intelligence, Secure Cloud Communication and Adaptive Quality Analytics**

ArXiv CS.AI 50%

arXiv:2607.20452v1 Announce Type: new Abstract: Modern software quality assurance demands intelligent, autonomous systems capable of adaptive decision-making across distributed cloud environments. This paper presents AINTMA (Agentic Intelligent Test Management Architecture), a...

<https://arxiv.org/abs/2607.20452>

6 CONF

**SOAP, Muon, and Beyond: Pushing LLM Pretraining Scales**

ArXiv CS.AI 50%

arXiv:2607.20548v1 Announce Type: cross Abstract: Higher-order optimizers such as Muon and SOAP offer faster convergence than AdamW, but their computational cost and numerical stability challenges have limited adoption at scale. In this work, we adapt and enhance preconditioned...

<https://arxiv.org/abs/2607.20548>

7 CONF

**ClickGuard: Detecting and Spoiling Clickbait News with Informativeness Measures and Large Language Models**

ArXiv CS.AI 50%

arXiv:2607.20463v1 Announce Type: new Abstract: This paper presents an AI-driven browser extension that identifies clickbait to help users avoid misleading Internet articles. Moving beyond traditional detection, the application employs a hybrid machine learning architecture that...

<https://arxiv.org/abs/2607.20463>

8 CONF

**Stochastic Sampling is Epistemically Shallow: The Dimensionality Gap Between Temperature Variation and Model Diversity in LLMs**

ArXiv CS.AI 50%

arXiv:2607.20464v1 Announce Type: new Abstract: When a language model gives different answers on repeated runs, does that variation reveal what it does not know? Self-consistency turns the variation into a per-question uncertainty estimate via majority voting. But does the same...

<https://arxiv.org/abs/2607.20464>

9 CONF

**IssueTrojanBench: Benchmarking AI Coding Agents Against Malicious Issue Requests**

ArXiv CS.AI 50%

arXiv:2607.20759v1 Announce Type: cross Abstract: AI coding agents powered by LLMs are increasingly integrated into real-world software development, where they generate, edit, and execute code with autonomous access to local files and tools. Coding agents inherit security risks...

<https://arxiv.org/abs/2607.20759>

0 CONF

**DC-Leap: Training-Free Acceleration of dLLMs via Draft-Guided Contiguous Leaping Decoding**

ArXiv CS.AI 50%

arXiv:2607.20467v1 Announce Type: new Abstract: While parallel decoding is central to the efficiency of Diffusion Large Language Models (dLLMs), current strategies are often hindered by overly conservative confidence thresholds. These thresholds, necessitated by the joint...

<https://arxiv.org/abs/2607.20467>

1 CONF

**DecodeShare: Tracing the Shared Subspace of LLM Decode-Time Decisions**

ArXiv CS.AI 50%

arXiv:2607.20469v1 Announce Type: new Abstract: Large language models (LLMs) handle many tasks with one set of parameters, but under KV-cached inference it is unclear what task-general structure, if any, is used at decode time rather than during prefill. We propose DecodeShare,...

<https://arxiv.org/abs/2607.20469>

2 CONF

**Robust Critics: Defending LLMs Against Multi-Turn Attacks**

ArXiv CS.AI 50%

arXiv:2607.20472v1 Announce Type: new Abstract: When a user asks a language model something harmful, is it a genuine attack or a misunderstood but well-meaning question? This ambiguity is one of the central challenges of LLM safety. A model that assumes the worst harms...

<https://arxiv.org/abs/2607.20472>

3 CONF

**SonicSampler: Unified Tile-Aware Kernels for LLM Sampling and Speculative Verification**

ArXiv CS.AI 50%

arXiv:2607.20475v1 Announce Type: new Abstract: Sampling in LLM inference comprises a combinatorial set of logit processing, token selection, and verification operations for speculative decoding. However, existing implementations either accelerate only subsets of this pipeline,...

<https://arxiv.org/abs/2607.20475>

4 CONF

**Benchmarking Unlearning for Vision Transformers**

ArXiv CS.AI 50%

arXiv:2602.20114v2 Announce Type: replace-cross Abstract: Machine unlearning (MU) refers to the post-training capability to remove (the influence of) training examples that are incorrect, biased, or leak sensitive/private information. MU is now widely regarded as critical for...

<https://arxiv.org/abs/2602.20114>

5 CONF

**Beyond Liars' Bench: The Impact of Lie Typology, Depth, and Sparsity on Deception Detection in LLMs**

ArXiv CS.AI 50%

arXiv:2607.20479v1 Announce Type: new Abstract: Training probes to detect deceptive outputs from large language models is still an open problem. Recent work has demonstrated that detection probes fail especially in out-of-domain scenarios -- training on one type of lie does not...

<https://arxiv.org/abs/2607.20479>

6 CONF

**Enabling Scalable Topology Inference in Distribution Systems via Constrained Multi-Source Inference**

ArXiv CS.AI 50%

arXiv:2607.20480v1 Announce Type: new Abstract: Accurate distribution system topology is essential for outage localization, voltage analytics, and operation of distribution grids, yet maintaining reliable connectivity records remains challenging in practice due to heterogeneous...

<https://arxiv.org/abs/2607.20480>

7 CONF

**Routing Without Training: Controllable-Ratio LLM Offloading via Reliability Gating**

ArXiv CS.AI 50%

arXiv:2607.20481v1 Announce Type: new Abstract: Local-cloud collaboration is a practical way to deploy large language models under resource constraints, but existing methods often rely on trained routers or collaboration-aware finetuning that tie routing behavior to a particular...

<https://arxiv.org/abs/2607.20481>

8 CONF

**PersonaTrail: Benchmarking Personalized Web Agents through Browsing Trails**

ArXiv CS.AI 50%

arXiv:2607.20482v1 Announce Type: new Abstract: Recent advances in large language models have enabled web agents to autonomously execute complex tasks. In practice, users frequently provide underspecified instructions, requiring agents to infer the missing context from their raw...

<https://arxiv.org/abs/2607.20482>

9 CONF

**Tractable Hierarchical Control of Autoregressive Language Models**

ArXiv CS.AI 50%

arXiv:2607.20483v1 Announce Type: new Abstract: Constraining the generation of autoregressive large language models (LLMs) is an important component of integrating language models into formal systems. In the generation of code and data for tasks like program synthesis, ensuring...

<https://arxiv.org/abs/2607.20483>

10 CONF

**The Devil is in the Spectrum: Mitigating Representation Collapse in LLMs via Topologically Regularized Side-Path**

ArXiv CS.AI 50%

arXiv:2607.20484v1 Announce Type: new Abstract: Large Language Models (LLMs) are fundamentally limited by representation collapse, a bottleneck that severely degrades long-context performance. We identify that existing approaches risk drifting into one of two pathological...

<https://arxiv.org/abs/2607.20484>



1 CONF

**Expectation Alignment of Language Models for Real-World User Expectations**

ArXiv CS.AI 50%

arXiv:2607.20485v1 Announce Type: new Abstract: Large language models (LLMs) have demonstrated remarkable performance on standard benchmarks, yet it remains largely unexplored whether they truly meet user expectations. Existing evaluation approaches, relying on model heuristics,...

<https://arxiv.org/abs/2607.20485>

2 CONF

**OPTScientist: Multi-Agent Discovery of Typed Optimizer Programs for Transformer Pretraining**

ArXiv CS.AI 50%

arXiv:2607.20486v1 Announce Type: new Abstract: Designing optimizers for modern deep learning remains a challenging scientific problem, requiring the joint consideration of optimization geometry, state dynamics, numerical stability, implementation constraints, and empirical...

<https://arxiv.org/abs/2607.20486>

3 CONF

**Directional Hallucinations: Ideological Drift in News-Grounded LLM Question Answering**

ArXiv CS.AI 50%

arXiv:2607.20487v1 Announce Type: new Abstract: Large language models (LLMs) are increasingly used to answer questions about political information, including in election-adjacent information settings where factual errors and ideological distortions are high-stakes. We present a...

<https://arxiv.org/abs/2607.20487>

4 CONF

**Autonomous Topology Mutation: Safe Runtime Restructuring for Multi-Agent LLM Systems with Capability, State, and Shadow Invariants**

ArXiv CS.AI 50%

arXiv:2607.20488v1 Announce Type: new Abstract: Multi-agent LLM frameworks typically fix their team topology at boot time. When an individual agent becomes overloaded at runtime, for example by mixing too many action categories, accumulating tool errors, or queueing behind too...

<https://arxiv.org/abs/2607.20488>

5 CONF

**Isolating LLM Alignment from Regex: Zero Coverage and Metric-Dependent Divergence Under Adversarial Mutation**

ArXiv CS.AI 50%

arXiv:2607.20494v1 Announce Type: new Abstract: Production LLM applications commonly stack a regex filter in front of model-side alignment; prior work found no measurable coverage gain from adding a live Gemini backend behind an active regex filter. We ask whether that ceiling...

<https://arxiv.org/abs/2607.20494>

6 CONF

**AISE-Bench: A Full-Cycle Curated Benchmark for Information Seeking on Academic Knowledge Graphs**

ArXiv CS.AI 50%

arXiv:2607.20498v1 Announce Type: new Abstract: Large language models (LLMs) augmented with tools are emerging as autonomous agents capable of using Web engine, APIs, and code to solve complex, long-horizon tasks. Current tool-using benchmarks for information seeking on academic...

<https://arxiv.org/abs/2607.20498>

7 CONF

**ExecuGraph: A Multi-Agent, Execution-Grounded Framework for Reliable Backend Code Synthesis with Large Language Models**

ArXiv CS.AI 50%

arXiv:2607.20499v1 Announce Type: new Abstract: Large Language Models generate plausible backend code, but a single-pass paradigm provides no guarantee of correctness or runtime reliability. We present ExecuGraph, a multi-agent framework that places execution-based validation at...

<https://arxiv.org/abs/2607.20499>

8 CONF

**FlowEdit: Information-Theoretic Control of LLM Reasoning Flows for Ill-posed Problems Involving Conflicts**

ArXiv CS.AI 50%

arXiv:2607.20500v1 Announce Type: new Abstract: Large Language Models (LLMs) perform strongly on well-specified reasoning tasks with a feasible answer. However, problems encountered in the open world can become ill-posed due to inconsistent conditions, conflicting statements, or...

<https://arxiv.org/abs/2607.20500>

9 CONF

**Optimizing Hypergraph-Based RAG: Toward Better Fact Extraction and Chunk Retrieval**

ArXiv CS.AI 50%

arXiv:2607.20506v1 Announce Type: new Abstract: GraphRAG enables deeper reasoning by structuring knowledge as graphs but struggles with n-ary facts. HyperGraphRAG uses hypergraphs for richer semantics, improving accuracy, yet relies on error-prone LLM extraction and inefficient...

<https://arxiv.org/abs/2607.20506>

0 CONF

**Unlearning Under Imbalance: Benchmarking Fairness in Multimodal LLM Unlearning**

ArXiv CS.AI 50%

arXiv:2607.21300v1 Announce Type: cross Abstract: Machine unlearning has emerged as a tool for removing personal data from trained models to comply with recent AI regulations. To evaluate unlearning effectiveness in multimodal large language models (MLLMs), prior works fine-tune...

<https://arxiv.org/abs/2607.21300>

1 CONF

**Reliability-Aware LLM Alignment from Inconsistent Human Feedback**

ArXiv CS.AI 50%

arXiv:2607.20515v1 Announce Type: new Abstract: Reinforcement Learning from Human Feedback (RLHF) is critical for aligning Large Language Models (LLMs) with human preferences. However, its efficacy is often compromised by the inherent inconsistency and subjectivity of human...

<https://arxiv.org/abs/2607.20515>

2 CONF

**Fragile Preferences: A Deep Dive Into Order Effects in Large Language Models**

ArXiv CS.AI 50%

arXiv:2506.14092v4 Announce Type: replace Abstract: Large language models (LLMs) are increasingly deployed in decision-support systems for high-stakes domains such as hiring and university admissions, where choices often involve selecting among competing alternatives. While...

<https://arxiv.org/abs/2506.14092>

3 CONF

**Scientific exploration, collaboration and labor division in the large language model era**

ArXiv CS.AI 50%

arXiv:2607.20923v1 Announce Type: cross Abstract: Large language models (LLMs) have rapidly and significantly entered scientific workflows, but it remains unclear how their diffusion is associated with changes in scientists' strategies in research directions and team building....

<https://arxiv.org/abs/2607.20923>

4 CONF

**Autonomous disproofs of the sum-product conjecture over  $\mathbb{R}$  with GPT-5.5 Pro**

ArXiv CS.AI 50%

arXiv:2607.20525v1 Announce Type: new Abstract: OpenAI's recent disproof of the Erdős unit distance conjecture marked a milestone for AI in mathematics. It also inspired another breakthrough: a human disproof of the Erdős-Szemerédi sum-product conjecture over  $\mathbb{R}$ ...

<https://arxiv.org/abs/2607.20525>

5 CONF

**SciExplore: Evaluating Autonomous Agents from Scientific Navigation to Information Integration**

ArXiv CS.AI 50%

arXiv:2607.20926v1 Announce Type: new Abstract: Scientific research involves complex information-seeking and reasoning workflows across heterogeneous sources. However, existing benchmarks primarily emphasize general-domain retrieval or static scientific question answering, and...

<https://arxiv.org/abs/2607.20926>

6 CONF

**DynamicMCPBench: A Trace-Grounded, Effect-Scored Benchmark for LLM Agents over Live MCP Servers**

ArXiv CS.AI 50%

arXiv:2607.20531v1 Announce Type: new Abstract: Large language model (LLM) agents are increasingly deployed over Model Context Protocol (MCP) servers, yet the benchmarks used to evaluate them score the final answer or a fixed "ground-truth" list of tools, both of which are...

<https://arxiv.org/abs/2607.20531>

7 CONF

**CMI-Mem: Toward Generalizable Long-Term Memory Management via CMI-Augmented Reinforcement Learning**

ArXiv CS.AI 50%

arXiv:2607.20553v1 Announce Type: new Abstract: Memory Manager models are pivotal in agent systems. Existing methods rely predominantly on LLM-judged synthetic question-answer (QA) pairs, making memory valuation dependent on sampled queries and the downstream reader. To address...

<https://arxiv.org/abs/2607.20553>

8 CONF

**Demographically-Informed Heat-Mortality Risk Curves via Risk Graph Neural Networks**

ArXiv CS.AI 50%

arXiv:2607.21131v1 Announce Type: cross Abstract: Estimating heat-related mortality risk is a core task in environmental epidemiology, typically addressed with Distributed Lag Non-linear Models (DLNMs); interpretable exposure-response surfaces fitted to temperature-mortality...

<https://arxiv.org/abs/2607.21131>

9 CONF

**A Knowledge-Injection Framework for Zero-Shot Adaptation of LLMs to Delirium Prediction**

ArXiv CS.AI 50%

arXiv:2607.20453v1 Announce Type: cross Abstract: Large language models show promise for clinical prediction, but zero-shot performance on specialized tasks is limited by incomplete domain knowledge, especially for smaller locally deployable models. We present a lightweight...

<https://arxiv.org/abs/2607.20453>

0 CONF

**The Human-AI Substitution Principle: When will you be replaced by AI in your organization?**

ArXiv CS.AI 50%

arXiv:2607.20781v1 Announce Type: new Abstract: Artificial Intelligence (AI) is rapidly transforming organizations, raising a fundamental organizational and economic question: when will a human employee be replaced by AI? We present an analytical model for studying Human--AI...

<https://arxiv.org/abs/2607.20781>

1 CONF

**Can an AI System Be Creative? A Critical Perspective from Art and Engineering**

ArXiv CS.AI 50%

arXiv:2607.20796v1 Announce Type: new Abstract: This paper examines the question of whether artificial intelligence (AI) systems can be creative, approached from the dual perspective of a researcher trained in electrical engineering, pattern recognition, machine learning, and...

<https://arxiv.org/abs/2607.20796>

2 CONF

**Enhancing Explainable Cardiac Diagnosis with Guide-Grounded Multimodal LLMs**

ArXiv CS.AI 50%

arXiv:2607.20814v1 Announce Type: new Abstract: The electrocardiogram (ECG) is a cornerstone of cardiac assessment, yet clinical deployment of deep learning models remains constrained by limited interpretability and the hallucination risk of large language models (LLMs)...

<https://arxiv.org/abs/2607.20814>

3 CONF

**Auditing Evidence Use in Medical LLM Diagnosis**

ArXiv CS.AI 50%

arXiv:2607.20848v1 Announce Type: new Abstract: Medical LLMs are often evaluated by whether they select the correct diagnosis, but diagnostic accuracy alone does not show whether the model used the case evidence appropriately. We present a behavioral audit of evidence use in...

<https://arxiv.org/abs/2607.20848>

4 CONF

**StackingNet: Collective Inference Across Independent AI Foundation Models**

ArXiv CS.AI 50%

arXiv:2602.13792v3 Announce Type: replace Abstract: Artificial intelligence built on large foundation models has transformed language understanding, computer vision, and reasoning, yet these systems remain isolated and cannot readily share their capabilities. Coordinating the...

<https://arxiv.org/abs/2602.13792>

- 5
CONF

### Traceable Scholarship: Page Anchors and Ariadne's Thread for Humanistic Inquiry in the Age of Generative AI

ArXiv CS.AI 50%

arXiv:2607.20916v1 Announce Type: new Abstract: Generative AI lets large language models produce scholarly-looking text within seconds, yet fluency does not equal valid explanation. The deepest risk is not factual error alone but the appearance that an explanation is already...

<https://arxiv.org/abs/2607.20916>
- 6
CONF

### Representing Entity Importance in AI Knowledge Systems: A Dual-Signal Framework of Audience Evaluation and Structural Authority

ArXiv CS.AI 50%

arXiv:2607.20925v1 Announce Type: new Abstract: AI knowledge systems require representations of entity importance for retrieval, recommendation, evidence selection, and knowledge-intensive reasoning. Yet importance is often reduced to a single score derived from either human...

<https://arxiv.org/abs/2607.20925>
- 7
CONF

### Clustered Edge Intelligence: Beyond Just Convergence of Edge Computing and AI

ArXiv CS.AI 50%

arXiv:2607.20937v1 Announce Type: new Abstract: We are moving from an information age to the age of intelligence. A decade, or possibly less than that, data will not be the gold anymore rather the derived intelligence out of the data and the information we posses from the edge...

<https://arxiv.org/abs/2607.20937>
- 8
CONF

### Delivery, Not Storage: Cue-Anchored Working Memory as a Harness Property for Coding Agents

ArXiv CS.AI 50%

arXiv:2607.20972v1 Announce Type: new Abstract: Coding agents ship with one kind of memory: documents. Instruction files, plan artifacts, and auto-written memory directories are deliberately authored and deliberately retrieved: the agent must choose to write them and choose to...

<https://arxiv.org/abs/2607.20972>
- 9
CONF

### Beyond Independent Optimization: Compression, MoE Routing, and Quantization Interactions in Multimodal Edge Intelligence

ArXiv CS.AI 50%

arXiv:2607.20981v1 Announce Type: new Abstract: Efficient multimodal inference is increasingly constrained not only by model quality or FLOP count, but also by the cost of preserving, moving, routing, caching, and quantizing multimodal representations under latency, memory, and...

<https://arxiv.org/abs/2607.20981>
- 0
CONF

### GuardianAgentBench: Where Agents Fail and How to Guard Them

ArXiv CS.AI 50%

arXiv:2607.20982v1 Announce Type: new Abstract: As large language model agents increasingly operate autonomously with access to tools and external environments, ensuring their safe and reliable behavior becomes critical. We present GuardianAgentBench (GABench), a benchmark of...

<https://arxiv.org/abs/2607.20982>

- 1

CONF

**Reexamining zero-shot summarization: Empirical investigation of trustworthiness of LLM-summarizers**

ArXiv CS.AI

50%

arXiv:2607.21010v1 Announce Type: new Abstract: Zero-shot summarization using Large Language Models (LLMs) has significantly advanced the abstractive summarization task by producing coherent and fluent summaries. However, underlying stochasticity of the large language models...

<https://arxiv.org/abs/2607.21010>
- 2

CONF

**Drive As You Like: Multi-Head Diffusion with Reinforcement Learning for Personalized Driving**

ArXiv CS.AI

50%

arXiv:2508.16947v2 Announce Type: replace-cross Abstract: Despite significant progress, imitation learning-based autonomous driving planners remain largely restricted to reproducing high-frequency biased behaviors, overlooking the inherent behavioral diversity of human driving....

<https://arxiv.org/abs/2508.16947>
- 3

CONF

**V-DEAL: Diagnosing Video Safety De-Calibration as an Understanding-Refusal Coupling Failure**

ArXiv CS.AI

50%

arXiv:2607.21151v1 Announce Type: new Abstract: As Video Large Language Models are increasingly deployed in real-world applications, ensuring their safety alignment has become critical. Counterintuitively, we find that harmful videos paired with benign queries achieve higher...

<https://arxiv.org/abs/2607.21151>
- 4

CONF

**SafeStep: AI-powered Travel Assistance for Elderly People with Frailty or Dementia**

ArXiv CS.AI

50%

arXiv:2607.21156v1 Announce Type: new Abstract: More than a million people in the UK suffer from frailty or dementia, which severely compromise their ability to travel in urban environments. This paper presents SafeStep, an AI-driven travel system that assists elderly users with...

<https://arxiv.org/abs/2607.21156>
- 5

CONF

**Safeguards for Speech2Speech LLM-Assistants: A Case Study in Automotive Applications**

ArXiv CS.AI

50%

arXiv:2607.21180v1 Announce Type: new Abstract: Recent advances have introduced speech-to-speech (S2S) conversational assistants capable of producing natural-sounding interactions, including non-verbal cues like tonality and mood. In the automotive domain, this enables intuitive...

<https://arxiv.org/abs/2607.21180>
- 6

CONF

**Explaining Weather Bulletins via ILP**

ArXiv CS.AI

50%

arXiv:2607.21184v1 Announce Type: new Abstract: Inductive Logic Programming (ILP) originated within the Logic Programming community in the Nineties as a framework for combining symbolic learning with declarative knowledge representation. Nowadays, mature ILP frameworks exist and...

<https://arxiv.org/abs/2607.21184>

- 7
CONF

### Identifying Good Rules for Efficient SAT Encodings of Single-Constant Multiplication Using Machine Learning

ArXiv CS.AI 50%

arXiv:2607.21188v1 Announce Type: new Abstract: The Single Constant Multiplication problem is a fundamental NP-hard optimization task in hardware design, which seeks to decompose a fixed constant using only additions, subtractions, and bit-shifts. Although dynamic programming...

<https://arxiv.org/abs/2607.21188>
- 8
CONF

### Bound-Founded Semantics for Answer Set Programming with Difference Constraints: Preliminary Report

ArXiv CS.AI 50%

arXiv:2607.21201v1 Announce Type: new Abstract: While the integration of linear constraints has significantly expanded the reach of Answer Set Programming (ASP), existing hybrid solvers often rely on disparate semantic underpinnings that lack a unified logical foundation. We...

<https://arxiv.org/abs/2607.21201>
- 9
CONF

### An LLM-Driven Workflow for Automated Process Control Strategy Generation and Tuning from Dynamic Process Models

ArXiv CS.AI 50%

arXiv:2607.21292v1 Announce Type: new Abstract: We present a structured large-language-model-driven workflow for automated multi-variable control design from dynamic process models. The workflow decomposes the design task into constrained code-generation steps: plant-interface...

<https://arxiv.org/abs/2607.21292>
- 0
CONF

### SPORD: A Simulation-Propose-then-OR-Dispose Approach for Supply Chain Planning

ArXiv CS.AI 50%

arXiv:2607.21354v1 Announce Type: new Abstract: For years, supply chain planning at e-commerce firms has operated as a collection of isolated projects. Each planning task from static network planning to dynamic warehouse assortment planning requires analysts to spend weeks...

<https://arxiv.org/abs/2607.21354>
- 1
CONF

### Towards Faithful Graph Explanations with Synergistic Edge Effects via Granular Balls

ArXiv CS.AI 50%

arXiv:2607.21381v1 Announce Type: new Abstract: Instance-level explanations aim to reveal the rationale behind a model's decisions for a specific graph. Previous methods explain graph neural networks (GNNs) by selecting important edges to induce subgraphs, where edge importance...

<https://arxiv.org/abs/2607.21381>
- 2
CONF

### Bridging the Gap Between Plausibility and Admissibility: Constraint-Aware Flow Maps for Dynamic Graph Systems

ArXiv CS.AI 50%

arXiv:2607.21421v1 Announce Type: new Abstract: Generative models can support decision-making under uncertainty by producing ensembles of plausible future system trajectories, but statistical plausibility does not ensure structural feasibility. This study investigates whether...

<https://arxiv.org/abs/2607.21421>

3 **CONF****Detecting LLM-Generated Tokens in Human--LLM Coauthored Text****ArXiv CS.AI 50%**

arXiv:2607.21458v1 Announce Type: new Abstract: The rise of human-AI collaborative writing has created a growing need for fine-grained detection methods that support localizing likely LLM-generated content in mixed-authorship documents. Existing methods for detecting...

<https://arxiv.org/abs/2607.21458>

4 **CONF****Toward Continuous Assurance for the Democratization of AI Agent Creation in Industry****ArXiv CS.AI 50%**

arXiv:2607.21495v1 Announce Type: new Abstract: AI agents are increasingly created inside organizations by non-engineering users through low-code, no-code, and conversational development environments. This democratization enables rapid local innovation, but it also creates a...

<https://arxiv.org/abs/2607.21495>

5 **CONF****Beyond Sycophancy: Structured Resistance and Compliance in LLM Moral Reasoning****ArXiv CS.AI 50%**

arXiv:2607.21558v1 Announce Type: new Abstract: Building socially calibrated large language models, which can learn from others without simply yielding to them, requires more than reducing sycophancy as a one-dimensional failure mode. Models must distinguish when to incorporate...

<https://arxiv.org/abs/2607.21558>

6 **CONF****From Attention to Frequency: Integration of Vision Transformer and FFT-ReLU for Enhanced Image Deblurring****ArXiv CS.AI 50%**

arXiv:2511.10806v1 Announce Type: cross Abstract: Image deblurring is vital in computer vision, aiming to recover sharp images from blurry ones caused by motion or camera shake. While deep learning approaches such as CNNs and Vision Transformers (ViTs) have advanced this field,...

<https://arxiv.org/abs/2511.10806>

7 **CONF****Knowledge Injection Exists in MoE? Exploring Expert-Aware Contrast Decoding in MoE for Mitigating LLMs'Hallucinations****ArXiv CS.AI 50%**

arXiv:2607.20426v1 Announce Type: cross Abstract: Existing LLM hallucination mitigation methods, including prompt engineering and model optimization, either hardly alter models'internal knowledge or have poor cross-domain generalization. Contrastive decoding mitigates...

<https://arxiv.org/abs/2607.20426>

8 **CONF****Is MoE Routing a Huffman Code? Discovering the Frequency-Diversity Law in Chain-of-Thought****ArXiv CS.AI 50%**

arXiv:2607.20427v1 Announce Type: cross Abstract: Mixture-of-Experts architectures have revolutionized scaling, yet the underlying logic of their routing remains a black box. In this paper, we uncover a fundamental governing principle: MoE routing is not merely selection, but a...

<https://arxiv.org/abs/2607.20427>



9 CONF

**More Is Not More: What Matters for Diversity in LLM Opinions?**

ArXiv CS.AI 50%

arXiv:2607.20429v1 Announce Type: cross Abstract: Large language models are increasingly used to simulate diverse human opinions in open-ended tasks such as synthetic surveys, focus group modeling, and public opinion prediction. However, LLM outputs exhibit systematic opinion...

<https://arxiv.org/abs/2607.20429>

0 CONF

**LLM-INSTRUCT at UZH Shared Task 2026: Constraint-Aware Retrieval and Selective Debate for Paragraph-Level Argument Mining**

ArXiv CS.AI 50%

arXiv:2607.20430v1 Announce Type: cross Abstract: We present LLM-INSTRUCT, the winning system for the UZH Shared Task at ArgMining 2026 on paragraph-level argument mining in UN and UNESCO resolutions. The task requires paragraph-type classification, prediction of a subset of 141...

<https://arxiv.org/abs/2607.20430>

1 CONF

**Moir: Let the Model Direct Its Own Story for Robust Cross-Domain Knowledge Editing**

ArXiv CS.AI 50%

arXiv:2607.20433v1 Announce Type: cross Abstract: While language models remain frozen at their training state, the world evolves continuously. Knowledge editing has emerged as a key alternative to full retraining, but its deployment is bottlenecked by the erosion of core...

<https://arxiv.org/abs/2607.20433>

2 CONF

**The Storyteller in the Model: Narrative Pattern Inheritance, Escalation Dynamics, and Alignment Governance in LLMs**

ArXiv CS.AI 50%

arXiv:2607.20449v1 Announce Type: cross Abstract: LLMs are trained predominantly on human-authored text, yet the structural and narrative conventions embedded in that text are rarely examined as a source of systematic behavioral influence, or as a governance risk in deployed...

<https://arxiv.org/abs/2607.20449>

3 CONF

**Response drift across frontier large language models**

ArXiv CS.AI 50%

arXiv:2607.20454v1 Announce Type: cross Abstract: All frontier large language models (LLMs) exhibit response drift -- producing outputs that deviate from expert-validated references -- yet the magnitude and structure of this drift remain uncharacterised by systematic human...

<https://arxiv.org/abs/2607.20454>

4 CONF

**Learn2Zinc: Fine-tuning Small Language Models for Text-to-Model Translation in MiniZinc**

ArXiv CS.AI 50%

arXiv:2607.20456v1 Announce Type: cross Abstract: Large language models excel at code generation for mainstream programming languages but struggle with rare, domain-specific languages such as MiniZinc, a constraint modeling language for combinatorial problems. We investigate...

<https://arxiv.org/abs/2607.20456>

5 CONF

**CAMeR: Keyword-Gated Hybrid Activation for Adaptive Memory Retention in LLM Agents**

ArXiv CS.AI 50%

arXiv:2607.20458v1 Announce Type: cross Abstract: Large language model (LLM) agents operating over extended dialogues accumulate vast amounts of information, yet existing memory systems either retain everything indiscriminately or apply uniform forgetting heuristics that fail to...

<https://arxiv.org/abs/2607.20458>

6 CONF

**Verifier-First Evaluation of Agentic LLMs for Infrastructure-as-Code Generation**

ArXiv CS.AI 50%

arXiv:2607.20478v1 Announce Type: cross Abstract: Infrastructure-as-Code (IaC) generation from natural language requires satisfying provider schemas, dependency planning, and organizational policy constraints, not merely producing syntactically plausible configurations. We...

<https://arxiv.org/abs/2607.20478>

7 CONF

**Scaling Closed-Loop Feature Channel Configuration with LLMs**

ArXiv CS.AI 50%

arXiv:2607.20516v1 Announce Type: cross Abstract: Promising initial results in closed-loop large-language-model-based channel-configuration search demonstrated that neural-network widths can be optimized directly through executable code generation and accuracy feedback. However,...

<https://arxiv.org/abs/2607.20516>

8 CONF

**Uncertainty-Aware Trust Estimation for Multi-LLM Systems via Structured Expert Judgement**

ArXiv CS.AI 50%

arXiv:2607.20529v1 Announce Type: cross Abstract: Large Language Model (LLM) ensembles are increasingly used to improve reliability by combining predictions from multiple LLMs. However, existing aggregation methods typically assume that all models are equally trustworthy,...

<https://arxiv.org/abs/2607.20529>

9 CONF

**Position: Stop Reactively Patching Your Model Every Time and Start Proactive Test-Driven AI Development**

ArXiv CS.AI 50%

arXiv:2607.20532v1 Announce Type: cross Abstract: Many modern AI systems are designed to operate under diverse, open-ended, use-cases. To help generalize deployed systems, many deployed-system maintenance pipelines use a reactive AI flywheel that observes emerging feedback from...

<https://arxiv.org/abs/2607.20532>

00 CONF

**A Graph Neural Network approach to zero-shot Digital Twins**

ArXiv CS.AI 50%

arXiv:2607.20535v1 Announce Type: cross Abstract: Traditional Predictive Digital Twins often remain geometrically rigid, requiring extensive retraining or fine-tuning whenever the underlying physical domain or boundary conditions change. To overcome this limitation, we present a...

<https://arxiv.org/abs/2607.20535>

01 CONF

**Codec-Gauge: Learning Compression-Friendly Gauges for Transformer KV Caches**

ArXiv CS.AI 50%

arXiv:2607.20538v1 Announce Type: cross Abstract: Long-context Transformer inference increasingly relies on KV-cache compression or quantization. Prior rotation and transform-coding results suggest that the channel basis of each key/value vector affects how faithfully a fixed...

<https://arxiv.org/abs/2607.20538>

02 CONF

**Leveraging Biokinetic Knowledge Priors for Data-Scarce Bioprocess Modeling**

ArXiv CS.AI 50%

arXiv:2607.20539v1 Announce Type: cross Abstract: While deep learning has accelerated drug discovery, its impact on biomanufacturing has been considerably more limited. The reason is data scarcity. Bioreactor experiments are high-cost, take days to weeks, and are rarely shared...

<https://arxiv.org/abs/2607.20539>

03 CONF

**From Noise to Diversity: Random Embedding Injection in LLM Reasoning**

ArXiv CS.AI 50%

arXiv:2605.11936v2 Announce Type: replace Abstract: Recent soft prompt research has tried to improve reasoning by inserting trained vectors into LLM inputs, yet whether the gain comes from the learned content or from the act of injection itself has not been carefully separated....

<https://arxiv.org/abs/2605.11936>

04 CONF

**Beyond SBDD: Geometric Deep Learning in Polypharmacology and Multi-target Drug Design**

ArXiv CS.AI 50%

arXiv:2607.20550v1 Announce Type: cross Abstract: The traditional "one drug, one target" paradigm of structure-based drug design (SBDD) frequently proves inadequate for treating multifactorial diseases such as cancer and neurodegenerative disorders, owing to compensatory...

<https://arxiv.org/abs/2607.20550>

05 CONF

**SenCos-GEM: SENet-Calibrated and Law-of-Cosines-Constrained Geometry-Enhanced Molecular Representation for Property Prediction**

ArXiv CS.AI 50%

arXiv:2607.20551v1 Announce Type: cross Abstract: Effective molecular representation learning is crucial for accurate molecular property prediction. Recently, numerous self-supervised learning (SSL) approaches leveraging 3D GNNs have been developed to capture comprehensive 3D...

<https://arxiv.org/abs/2607.20551>

06 CONF

**Monkey King Bang: A Unified Scientific Multimodal Foundation Model**

ArXiv CS.AI 50%

arXiv:2607.20557v1 Announce Type: cross Abstract: Scientific discovery is increasingly shifting from isolated disciplines to multi-domain reasoning, and AI for science faces a similar transition. Existing systems are either specialised for individual domains or unify scientific...

<https://arxiv.org/abs/2607.20557>

07 CONF

**Geometric Configurations of Perturbed Jailbreak Prompts**

ArXiv CS.AI 50%

arXiv:2607.20581v1 Announce Type: cross Abstract: Perturbation techniques that turn unsuccessful jailbreak prompts into successful ones are continuously evolving, constituting a major security threat to LLM safety. In this paper, we investigate the internal representations of...

<https://arxiv.org/abs/2607.20581>

08 CONF

**Cycle-Consistent and Uncertainty-Aware Neural Surrogates for Tokamak Edge Plasmas**

ArXiv CS.AI 50%

arXiv:2607.21407v1 Announce Type: cross Abstract: The boundary and divertor plasma govern how a tokamak exhausts power and particles, setting heat fluxes, target conditions, and the onset of detachment. Predicting these quantities is essential for operating current and future...

<https://arxiv.org/abs/2607.21407>

09 CONF

**When Does Recurrence Become an Algorithm? Convergence Selection in Weight-Tied Looped Transformers**

ArXiv CS.AI 50%

arXiv:2607.20594v1 Announce Type: cross Abstract: When does a weight-tied looped transformer -- one block applied T times -- implement an actual algorithm? We answer with four findings from controlled populations on group word problems. (1) The budget law: free training installs...

<https://arxiv.org/abs/2607.20594>

10 CONF

**RealVDeblur: One-Step Diffusion for Generalizable Real-World Video Deblurring**

ArXiv CS.AI 50%

arXiv:2607.20628v1 Announce Type: cross Abstract: Real-world video deblurring remains challenging due to diverse motion patterns, complex degradations, and the scarcity of realistic training data, yet robust restoration is critical for downstream pipelines such as mobile imaging...

<https://arxiv.org/abs/2607.20628>

11 CONF

**Demonstrating GenDB: Instance-Optimized and Customized Query Processing Code Generation via LLM Agents**

ArXiv CS.AI 50%

arXiv:2607.20630v1 Announce Type: cross Abstract: Traditional query processing engines require continuous development and extensions to support new techniques and user requirements, and in some cases, entirely new systems must be built from scratch. However, these engines are...

<https://arxiv.org/abs/2607.20630>

12 CONF

**Interpretable Embeddings with Sparse Autoencoders: A Data Analysis Toolkit**

ArXiv CS.AI 50%

arXiv:2512.10092v2 Announce Type: replace Abstract: Analyzing large-scale text corpora is a core challenge in machine learning, crucial for tasks like identifying undesirable model behaviors or biases in training data. Current methods often rely on costly LLM-based techniques...

<https://arxiv.org/abs/2512.10092>

13 CONF

**DynaMark: A Reinforcement Learning Framework for Dynamic Watermarking in Industrial Machine Tool Controllers**

ArXiv CS.AI 50%

arXiv:2508.21797v2 Announce Type: replace-cross Abstract: Industry 4.0's highly networked Machine Tool Controllers (MTCs) are prime targets for replay attacks that use outdated sensor data to manipulate actuators. Dynamic watermarking can reveal such tampering, but current...

<https://arxiv.org/abs/2508.21797>

14 CONF

**From Agent Failures to Text Policies: What Works and What Breaks**

ArXiv CS.AI 50%

arXiv:2607.20668v1 Announce Type: cross Abstract: TextGrad improves language-model systems by revising text from feedback. Its core thesis is that natural-language feedback can act as a gradient for optimizing text components without changing model weights. Applying it to agents...

<https://arxiv.org/abs/2607.20668>

15 CONF

**A Framework for Reputation Aware Uninorm-driven Consensus Algorithms for Blockchain Networks**

ArXiv CS.AI 50%

arXiv:2607.20700v1 Announce Type: cross Abstract: The operation of blockchain is governed by consensus algorithms (CA). Several consensus mechanisms require significant computational power, while others necessitate high amounts of stakes to select the participant to validate and...

<https://arxiv.org/abs/2607.20700>

16 CONF

**Are Diversity Metrics Measuring Diversity? A Capability-Controlled Audit of Majority-Vote Gain in LLM Ensembles**

ArXiv CS.AI 50%

arXiv:2607.20768v1 Announce Type: cross Abstract: Majority voting over LLMs is widely assumed to benefit from diversity, and diversity measures are used to choose which models to combine. We ask whether five such measures track diversity or mainly re-express capability, auditing...

<https://arxiv.org/abs/2607.20768>

17 CONF

**HARP: The Human--AI Research Platform**

ArXiv CS.AI 50%

arXiv:2607.20773v1 Announce Type: cross Abstract: Large language models (LLMs) have shifted human--computer interaction from "traditional" interface journeys toward more conversational exchanges. Researchers studying HCI and UI use moderated usability sessions, interviews,...

<https://arxiv.org/abs/2607.20773>

18 CONF

**Beyond Heavy Log Curation: Perplexity-Based APT Detection via Unsupervised, Context-Augmented Language Models**

ArXiv CS.AI 50%

arXiv:2607.20832v1 Announce Type: cross Abstract: Advanced Persistent Threats (APTs) remain difficult to detect because only a small fraction of events in large-scale logs are attack-related, and investigation is expensive and hard to scale. Prior machine-learning approaches can...

<https://arxiv.org/abs/2607.20832>

19 CONF

**Glucotune: A Unified Framework for Blood Glucose Preprocessing, Forecasting, and Benchmarking in Diabetes**

ArXiv CS.AI 50%

arXiv:2607.21117v1 Announce Type: cross Abstract: Preprocessing blood glucose time-series data is a critical yet often overlooked step in developing data-driven methods for diabetes management, particularly for type 1 diabetes. The lack of standardized preprocessing workflows...

<https://arxiv.org/abs/2607.21117>

20 CONF

**Hardware-Software Co-Design for Float16 On-Device Training on RISC-V Single-Core**

ArXiv CS.AI 50%

arXiv:2607.21130v1 Announce Type: cross Abstract: By leveraging standard RISC-V extensions, namely Zfh (scalar float16) and Zvfh (vector float16), this work proposes an open-source framework to enable complete on-device training on resource-constrained RISC-V single-core. Our...

<https://arxiv.org/abs/2607.21130>

21 CONF

**One More Turn, Less Regret: A Regret-Based Multi-Turn Benchmark for LLMs' Clarification Policies**

ArXiv CS.AI 50%

arXiv:2607.21143v1 Announce Type: cross Abstract: Ambiguous user requests make clarification a sequential decision problem for conversational LLM assistants: they must decide whether to ask, what to ask, when to stop, and when to answer. We introduce RegretBench, a multi-turn...

<https://arxiv.org/abs/2607.21143>

22 CONF

**CRAG-MM-Diagnostics: Enabling Stage-Wise Analysis of Knowledge-Intensive VQA**

ArXiv CS.AI 50%

arXiv:2607.21155v1 Announce Type: cross Abstract: Knowledge-Intensive Visual Question Answering (KI-VQA) benchmarks evaluate Vision-Language Models (VLMs) as multimodal knowledge assistants by requiring external information beyond a provided image to answer questions. KI-VQA...

<https://arxiv.org/abs/2607.21155>

23 CONF

**Case study: proving  $\sqrt{2}$  irrational with LPTP and an LLM**

ArXiv CS.AI 50%

arXiv:2607.21187v1 Announce Type: cross Abstract: We present the interactions with an LLM (Large Language Model) aiming at proving that the square root of 2 is not a rational number in an LP (Logic Programming) context. We start from a few basic pure logic programming predicate...

<https://arxiv.org/abs/2607.21187>

24 CONF

**pAI-Econ-claude: A Gated Human-in-the-Loop Multi-Agent Architecture for AI-Assisted Economic Theory Development**

ArXiv CS.AI 50%

arXiv:2607.21268v1 Announce Type: cross Abstract: In many social-science research tasks, such as economics, LLM-based agents must produce outputs for which no cheap, task-complete, machine-readable correctness signal exists. This creates a distinctive reliability problem for...

<https://arxiv.org/abs/2607.21268>

25 CONF

**A Comparative Evaluation of Embeddings and LLMs in a Greek Book Publisher Setting - The CUP Dataset**

ArXiv CS.AI 50%

arXiv:2607.21274v1 Announce Type: cross Abstract: We present CUP, a Greek book retrieval benchmark consisting of 868 catalog records and 104 expert-annotated queries with graded relevance judgments. We evaluate sparse (BM25), dense (sentence-transformers), hybrid, and...

<https://arxiv.org/abs/2607.21274>

26 CONF

**AI Assistants Overassist**

ArXiv CS.AI 50%

arXiv:2607.21306v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly used as tutors and thought partners, helping users reason through problems. While guidance from AI assistants can scaffold thinking and foster learning, such benefits depend on how...

<https://arxiv.org/abs/2607.21306>

27 CONF

**Scaling Up Formal Representation of Clinical Trial Protocols in Ensemble Logic Using LLMs: A Preliminary Study**

ArXiv CS.AI 50%

arXiv:2607.21307v1 Announce Type: cross Abstract: The reliance on unstructured free text for documenting clinical trial protocols creates a significant barrier to automated reasoning, cohort discovery, and trial simulation. The lack of formal structure obscures critical temporal...

<https://arxiv.org/abs/2607.21307>

28 CONF

**GRADRAG: Cross-Component Prompt Adaptation for Coordinated Multi-Agent RAG**

ArXiv CS.AI 50%

arXiv:2607.21324v1 Announce Type: cross Abstract: Retrieval-Augmented Generation (RAG) systems increasingly employ multiple LLM agents. Yet, most prior work optimizes components in isolation rather than coordinating improvements across the pipeline. We introduce GRADRAG, a...

<https://arxiv.org/abs/2607.21324>

29 CONF

**Toward cryptographically verifiable authorization for autonomous AI agents: A security hypothesis, preliminary formal model, and proof-of-concept implementation**

ArXiv CS.AI 50%

arXiv:2607.21325v1 Announce Type: cross Abstract: Autonomous AI agents increasingly execute actions, invoke tools, and operate on protected resources with limited human oversight. Existing authentication and authorization mechanisms establish identity and delegate authority, but...

<https://arxiv.org/abs/2607.21325>

30 CONF

**From Static Bibliometrics to Dynamic Knowledge Graphs: An LLM-Powered Framework for Modernizing Science, Technology, and Innovation (STI) Analytics**

ArXiv CS.AI 50%

arXiv:2607.21327v1 Announce Type: cross Abstract: Bibliometric indicators - citation counts, h-indexes, co-authorship networks - have long anchored science, technology, and innovation (STI) analytics, yet suffer from temporal lag, semantic shallowness, and an inability to...

<https://arxiv.org/abs/2607.21327>

31 CONF

**Phonetic forced alignment for low-resource language varieties: Model training and evaluation on Chengdu Mandarin**

ArXiv CS.AI 50%

arXiv:2607.21332v1 Announce Type: cross Abstract: Phonetic forced alignment is a key technique in phonetic research, yet existing alignment systems lack specialized models for low-resource language varieties. We address this by training text-dependent and text-independent...

<https://arxiv.org/abs/2607.21332>

32 CONF

**M<sup>3</sup>-Gen: Interpretable Multimodal Generation of Gene Expression Profiles Using Clinical and Imaging Data**

ArXiv CS.AI 50%

arXiv:2607.21343v1 Announce Type: cross Abstract: Integrating heterogeneous biomedical data, including clinical metadata, histopathology images, and molecular profiles, is crucial for comprehensive disease understanding. However, gene expression data acquisition remains...

<https://arxiv.org/abs/2607.21343>

33 CONF

**DINode: Continuous Vision-Text Alignment for Open-Vocabulary Semantic Segmentation**

ArXiv CS.AI 50%

arXiv:2607.21371v1 Announce Type: cross Abstract: Open-vocabulary semantic segmentation (OVSS) leverages textual semantics to segment objects beyond predefined categories. While the self-supervised model DINOv3 provides strong structured visual representations, its lack of...

<https://arxiv.org/abs/2607.21371>

34 CONF

**Mean-to-Score Discrete Diffusion: Posterior-Mean Denoisers for Score Entropy**

ArXiv CS.AI 50%

arXiv:2607.21372v1 Announce Type: cross Abstract: Score Entropy Discrete Diffusion (SEDD) parameterizes discrete reverse processes with unconstrained positive score ratios. While positivity guarantees nonnegative reverse jump rates, it does not ensure Bayes realizability: ratios...

<https://arxiv.org/abs/2607.21372>

35 CONF

**VoLN: Vision-Only Long-Horizon Navigation---Paradigm, Benchmark, and Method**

ArXiv CS.AI 50%

arXiv:2607.21400v1 Announce Type: cross Abstract: Vision-and-Language Navigation (VLN) enables embodied agents to follow natural-language instructions. However, route-level instructions commonly encode spatial priors, such as orientation, distance, and layout, that are not...

<https://arxiv.org/abs/2607.21400>

36 CONF

**When Are Reasoning-Based Guardrails Not Efficient? ResponseGuard: A Fast Vision-Language Guard for Real-Time Moderation**

ArXiv CS.AI 50%

arXiv:2607.21401v1 Announce Type: cross Abstract: A vision-language AI assistant returns its answer as a stream of generated tokens. Therefore, a safety guard that watches that answer has to keep up with the stream and stop a harmful reply before a user reads it. Recent...

<https://arxiv.org/abs/2607.21401>



37 CONF

**Token Budget Saturation and Mechanistic Early Detection of Reasoning Non-Convergence in Chain-of-Thought Models**

ArXiv CS.AI 50%

arXiv:2607.21433v1 Announce Type: cross Abstract: Chain-of-thought reasoning models such as DeepSeek-R1-Distill-Qwen-7B exhibit a bimodal convergence pattern: generations either terminate within a token budget (converged) or exhaust it without reaching a conclusion...

<https://arxiv.org/abs/2607.21433>

38 CONF

**Adaptive Identity Anchoring: Closed-Loop Keyframe Placement for Synthetic Paired Supervision in Video Face Swapping**

ArXiv CS.AI 50%

arXiv:2607.21434v1 Announce Type: cross Abstract: Video face swapping has no natural paired supervision: no real footage exists of one person's face performing another person's video. The strongest current answer, DreamID-V's SyncID-Pipe, mints pairs by replacing the identity in...

<https://arxiv.org/abs/2607.21434>

39 CONF

**RUMBA: Russian User Memory Benchmark**

ArXiv CS.AI 50%

arXiv:2607.21447v1 Announce Type: cross Abstract: The ability to handle long-term memory in LLMs is becoming increasingly critical, yet existing benchmarks remain English-centric and rely on aggregate retrieval metrics, failing to capture interactions between long-range context,...

<https://arxiv.org/abs/2607.21447>

40 CONF

**GS-Agent: Creating 4D Physical Worlds With Generative Simulation**

ArXiv CS.AI 50%

arXiv:2607.21522v1 Announce Type: cross Abstract: Creating dynamic and physically realistic 4D worlds from natural language descriptions is both fascinating and challenging. Traditional computer graphics methods rely on manual creation, requiring extensive human effort to...

<https://arxiv.org/abs/2607.21522>

41 CONF

**From Resource Flow to Executable Tests: Petri-Net-Guided LLM Test Generation for Concurrent Stateful Rust APIs**

ArXiv CS.AI 50%

arXiv:2607.21530v1 Announce Type: cross Abstract: Concurrent stateful library APIs expose behavior through evolving resource ownership, lifecycle states, and competing interleavings. Large language models can synthesize executable Rust tests, but their outputs often violate API...

<https://arxiv.org/abs/2607.21530>

42 CONF

**Barzilai-Borwein Fails Superlinear Convergence on an Open Set of Quadratics for Every Dimension  $n \geq 4$** 

ArXiv CS.AI 50%

arXiv:2607.21579v1 Announce Type: cross Abstract: Barzilai-Borwein (BB) method has shown strong practical performance in continuous optimization, yet its convergence dynamics remains poorly understood. In particular, a central unresolved question is whether BB converges...

<https://arxiv.org/abs/2607.21579>

43 CONF

**Generative Artificial Intelligence in Bioinformatics: A Systematic Review of Models, Applications, and Methodological Advances**

ArXiv CS.AI 50%

arXiv:2511.03354v2 Announce Type: replace-cross Abstract: Generative artificial intelligence (GenAI) is transforming bioinformatics by advancing genomics, proteomics, transcriptomics, structural biology, and drug discovery. Following the Preferred Reporting Items for Systematic...

<https://arxiv.org/abs/2511.03354>

44 CONF

**Diagnosing Pathological Chain-of-Thought in Reasoning Models**

ArXiv CS.AI 50%

arXiv:2602.13904v2 Announce Type: replace Abstract: Chain-of-thought (CoT) reasoning is fundamental to modern LLM architectures and represents a critical intervention point for AI safety. However, CoT reasoning may exhibit failure modes that we note as pathologies, which prevent...

<https://arxiv.org/abs/2602.13904>

45 CONF

**Post-Hoc Reasoning in Chain of Thought: Decoding and Steering Pre-Committed Answers**

ArXiv CS.AI 50%

arXiv:2603.01437v2 Announce Type: replace Abstract: As chain of thought (CoT) has become central to scaling reasoning capabilities in large language models (LLMs), it has also emerged as a promising tool for interpretability, suggesting the opportunity to understand model...

<https://arxiv.org/abs/2603.01437>

46 CONF

**Evaluating Risks in Weak-to-Strong Alignment: A Bias-Variance Perspective**

ArXiv CS.AI 50%

arXiv:2604.25077v2 Announce Type: replace Abstract: Weak-to-strong alignment offers a promising route to scalable supervision, but it can fail when a strong model becomes confidently wrong on examples that lie in the weak model's blind spots. Understanding such failures requires...

<https://arxiv.org/abs/2604.25077>

47 CONF

**A Three-Phase Foundation Model for Tax-Aware Personalized Portfolio Management**

ArXiv CS.AI 50%

arXiv:2606.30997v3 Announce Type: replace Abstract: We present a three-phase deep reinforcement learning system for personalized portfolio management that addresses three limitations shared by all prior financial RL work: 1) ticker lock-in, 2) monolithic objectives , and 3)...

<https://arxiv.org/abs/2606.30997>

48 CONF

**Generative AI and Agency in Education: A Critical Scoping Review and Thematic Analysis**

ArXiv CS.AI 50%

arXiv:2411.00631v2 Announce Type: replace-cross Abstract: This scoping review examines the relationship between Generative AI (GenAI) and agency in education, analyzing the literature available through the lens of Critical Digital Pedagogy. Following PRISMA-ScR guidelines, we...

<https://arxiv.org/abs/2411.00631>

49 **CONF****MELLA: Bridging Linguistic Capability and Cultural Groundedness for Low-Resource Language MLLMs**

ArXiv CS.AI 50%

arXiv:2508.05502v2 Announce Type: replace-cross Abstract: Multimodal Large Language Models (MLLMs) perform strongly in high-resource languages, yet often produce fluent but culturally "thin" descriptions in low-resource settings. We argue that this failure is not merely a...

<https://arxiv.org/abs/2508.05502>

50 **CONF****Equivariant Conditional Diffusion Model for Head and Neck CT Image Synthesis from CBCT**

ArXiv CS.AI 50%

arXiv:2509.21913v2 Announce Type: replace-cross Abstract: Background: Cone-beam computed tomography CBCT is a commonly used modality for image guided radiotherapy. It offers real time anatomical visualization with low acquisition cost and dose. Nevertheless, photon scattering...

<https://arxiv.org/abs/2509.21913>

51 **CONF****Backpropagation-Free Test-Time Adaptation for Lightweight EEG-Based Brain-Computer Interfaces**

ArXiv CS.AI 50%

arXiv:2601.07556v2 Announce Type: replace-cross Abstract: Electroencephalogram (EEG)-based brain-computer interfaces (BCIs) face significant deployment challenges due to inter-subject variability, signal non-stationarity, and computational constraints. While test-time adaptation...

<https://arxiv.org/abs/2601.07556>

52 **CONF****PILD: Physics-Informed Learning via Diffusion**

ArXiv CS.AI 50%

arXiv:2601.21284v2 Announce Type: replace-cross Abstract: Diffusion models have emerged as powerful generative tools for modeling complex data distributions, yet their purely data-driven nature limits applicability in engineering and scientific problems where physical laws must...

<https://arxiv.org/abs/2601.21284>

53 **CONF****Self-Evolving Recommendation System: End-To-End Autonomous Model Optimization With LLM Agents**

ArXiv CS.AI 50%

arXiv:2602.10226v2 Announce Type: replace-cross Abstract: Optimizing large-scale machine learning systems, such as recommendation models for global video platforms, requires navigating a massive hyperparameter search space and, more critically, designing sophisticated...

<https://arxiv.org/abs/2602.10226>

54 **CONF****SR-TTT Does Not Learn Retrieval: A Correction and Mechanistic Post-Mortem of Surprisal-Aware Residual Test-Time Training**

ArXiv CS.AI 50%

arXiv:2603.06642v2 Announce Type: replace-cross Abstract: Test-Time Training (TTT) language models replace the KV-cache with fast weights updated during inference, achieving O(1) memory but suffering catastrophic failure on exact-recall tasks. Version 1 of this work proposed...

<https://arxiv.org/abs/2603.06642>

55 CONF

**SafeHarbor: Defining Precise Decision Boundaries via Hierarchical Memory-Augmented Guardrail for LLM Agent Safety**

ArXiv CS.AI 50%

arXiv:2605.05704v3 Announce Type: replace-cross Abstract: Recent advances in foundation models have transformed LLMs from passive conversational systems into autonomous agents capable of reasoning and tool execution. While these capabilities unlock substantial practical value,...

<https://arxiv.org/abs/2605.05704>

56 CONF

**AI Security Policy Should Assess Systems, Not Only Models**

ArXiv CS.AI 50%

arXiv:2605.09504v2 Announce Type: replace-cross Abstract: We present swarm-attack, an open-source adversarial testing framework in which multiple lightweight LLM agents coordinate through shared memory, parallel exploration, and evolutionary optimization. Together, our results...

<https://arxiv.org/abs/2605.09504>

57 CONF

**Riemannian Deep Learning: Modules, Networks, and Geometries**

ArXiv CS.AI 50%

arXiv:2607.19305v2 Announce Type: replace-cross Abstract: Deep neural networks on manifold-valued representations have attracted growing interest, but many basic components remain tied to specific manifolds, rely on Euclidean approximations, or require costly and numerically...

<https://arxiv.org/abs/2607.19305>